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B.Tech. Degree VII Semester Supplementary Examination
April 2019

ME 15-1701 REFRIGERATION AND AIR CONDITIONING
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain the need of refrigeration system in an aircraft.
- (b) What is the principle behind vortex tube refrigeration?
- (c) Why Carnot refrigeration cycle is considered as a theoretical cycle?
- (d) Why in some cases multistage refrigeration system is preferred?
- (e) What is a flash chamber? What is its use?
- (f) How low pressure is maintained in the evaporator in electrolux absorption refrigerator?
- (g) Discuss the basic principle behind the working of capillary tube as an expansion device. What are its advantages?
- (h) What do you mean by clearance factor of a compressor? How it varies with volumetric efficiency?
- (i) Define: (i) relative humidity (ii) humidity ratio.
- (j) What is by pass factor?



PART B

(4 × 10 = 40)

- II. (a) A reversed Carnot cycle delivers 60000 kJ/min to a conditioned space when used as a heat pump. If the space is maintained at 25°C when the ambient temperature is 10°C, find: (i) amount of heat drawn from the surroundings (ii) power required to run the pump (iii) if the required heat is supplied by electric heaters determine the power consumed. (5)
- (b) Derive the expression for CoP for the reversed Brayton cycle, assuming polytropic expansion and compression. (5)

OR

- III. (a) An air refrigerator system operating on reversed Brayton cycle takes in air from cold room at -6°C and compresses it from 1.04 bar to 6.2 bar. The index of compression is 1.28. The compressed air is cooled to 25°C. The ambient temperature is 18°C. Air expands in an expander where index of compression is 1.38. Determine: (5)
 - (i) CoP of the system.
 - (ii) Quantity of air circulated per minute for production of 1500 kg of ice per day at 0°C from water at 18°C.
 - (iii) Capacity of the plant.
- (b) Discuss simple air refrigeration system and boot strap air refrigeration system. Which one is more suitable for low flight speeds? Why? (5)
- IV. (a) An NH₃ refrigerator produces 30 tonnes of ice from and at 0°C in a day. The temperature range in the compressor is 26°C to -14°C. The vapor is dry saturated at the end of compression. Assuming CoP as 60% of theoretical value, calculate the power required to drive the compressor. Latent heat of ice = 335 kJ/kg. Property values are as follows: (5)

(P.T.O.)

Temperature (°C)	Enthalpy-vapour (kJ/kg)	Enthalpy-liquid (kJ/kg)	Entropy-liquid (kJ/kgK)	Entropy vapour (kJ/kgK)
26	1483.72	322.73	1.425	5.306
-14	1445.47	135.82	0.759	4.813

- (b) With a neat sketch explain Lithium Bromide vapour absorption refrigeration system. (5)

OR

- V. (a) What are the desirable properties of a refrigerant? Why introduction of newer refrigerants is considered as the need of the hour. (5)
- (b) A refrigeration machine is required to produce ice at 0°C from water at 20°C. The machine has a condenser temperature of 298 K while the evaporator temperature is 268 K. The relative efficiency of the machine is 50% and 6 kg of refrigerant is circulated through the system per minute. The refrigerant enters the compressor with dryness fraction 0.6 and leaves as dry saturated vapour. Specific heat of water is 4.187 kJ/kgK and latent heat of ice 335 kJ/kg. Calculate the amount of ice produced in 24 hrs. The properties of the refrigerant are as follows: (5)

Temperature (K)	Liquid heat (kJ/kg)	Latent heat (kJ/kg)	Entropy of liquid (kJ/kgK)
298	59.7	138	0.2232
268	31.4	154	0.1251

- VI. Discuss the different types of condensers and evaporators used for refrigeration. (10)

OR

- VII. (a) With a neat sketch explain any two rotary compressors. (5)
- (b) Define volumetric efficiency of a compressor. What are the factors influencing volumetric efficiency? (5)
- VIII. (a) Atmospheric air at 15°C DBT and 30% RH passes through a furnace and through a humidifier, in such a way that the final DBT is 32°C and 40% RH. Determine (i) heat and moisture added to air (ii) sensible heat factor of the process. (5)
- (b) With the support of diagrams, briefly explain different psychrometric processes. (5)

OR

- IX. The following data relate to an air conditioned room: (10)

Outside conditions	36°C DBT, 27°C WBT
Inside conditions	24°C DBT, 50% RH
Sensible heat load	12 kW
Latent heat load	7.3 kW
Apparatus Dew point temperature	7°C
Bypass factor of the cooling coil used	0.1

Return air from the room is mixed with the outside air before entering the cooling coil in the ratio of 4:1 and return air from room is also mixed after the cooling coil in the ratio 1:4. The air may be reheated, if necessary before supplying to the conditioned room. Find:

- (i) Supply air condition to the room
- (ii) Refrigeration load
- (iii) Quantity of fresh air supplied
